Improving Deep Neural Networks

Optimization · Regularization · Tuning · Transfer Learning

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Course Outline

Module 1: Optimization

- Gradient Descent Variants
- Momentum & RMSProp
- Adam Optimizer
- Learning Rate Scheduling

Module 2: Regularization

- Dropout
- Batch Normalization
- L1 / L2 Regularization

Module 3: Tuning & Transfer

- Hyperparameter Tuning
- Transfer Learning
- Fine-Tuning Strategies

Module 4: Generalization

- Bias–Variance Tradeoff
- Overfitting Strategies
- Small Dataset Techniques
- Data Augmentation

Optimization

Optimization

Making training fast, stable, and effective

Why Optimization Matters

- Neural networks learn by minimizing a **loss function** $\mathcal{L}(\theta)$
- **Challenge:** High-dimensional, non-convex loss landscapes
- Vanilla stochastic gradient descent (SGD) is **slow and unstable**

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \text{:- Vanilla SGD update}$$

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  - Saddle points and plateaus
  - Noisy gradient estimates
  - Ill-conditioned curvature

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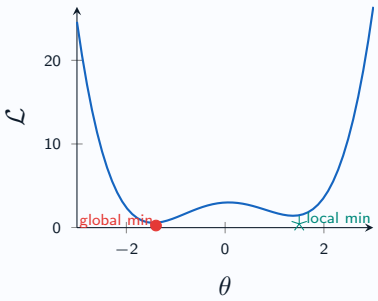
Loss Landscape



SGD Variants

Batch GD

- Uses **all** training samples
- Stable but very slow
- Not practical for large data

$\nabla \mathcal{L} = \frac{1}{m} \sum_{i=1}^{m} \nabla \ell_i$

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- Balance of speed & stability
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Best Practice

Typical batch sizes: 32, 64, 128, 256. Larger batches $\Rightarrow$ more stable gradients but need tuned LR and more memory.

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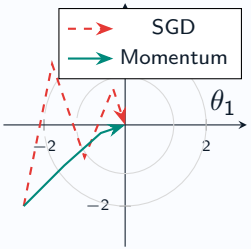
- $\beta \approx 0.9$ is the momentum coefficient
- Accumulates gradients like a ball rolling downhill

Momentum vs SGD Path



RMSProp

Key Idea: Adapt the learning rate *per parameter* by scaling it with the **root mean square of past gradients**.

$$s_t = \beta_2 \, s_{t-1} + (1 - \beta_2) \, g_t^2$$
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Benefits:

- Larger LR for infrequent features
- Smaller LR for frequent features
- Works well with non-stationary objectives
- Typical: $\beta_2 = 0.999$, $\epsilon = 10^{-8}$

Comparison: SGD vs RMSProp

	SGD	RMSProp
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Note

RMSProp lacks theoretical convergence guarantees in general non-convex settings.

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- ✅ Computationally efficient
- ✅ Works with sparse gradients
- ✅ Little memory requirement
- ✅ Invariant to gradient scaling
- ⚠️ May generalize worse than SGD+momentum in some tasks

Learning Rate: The Most Critical Hyperparameter

- Controls the **step size** of each update
- **Too large** $\Rightarrow$ divergence / oscillation
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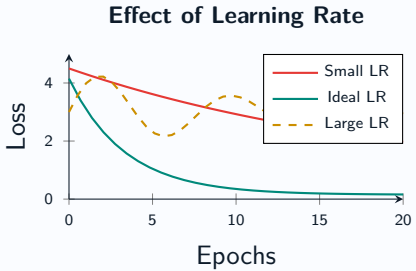
- **Just right** $\Rightarrow$ fast, stable learning



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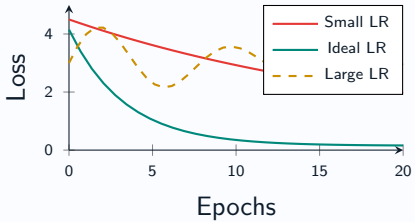

LR Range Test

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Plot loss vs LR. Choose LR just **before** the loss stops falling.

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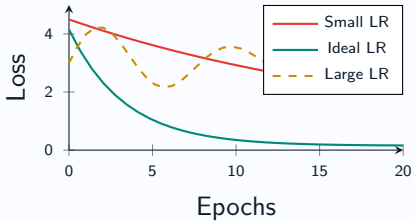

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Rule of thumb:

LR $\approx 10^{-3}$ for Adam

LR $\approx 10^{-2}$ for SGD

Regularization

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Preventing overfitting and improving generalization

Bias–Variance Tradeoff

$$\mathbb{E}[\text{Error}^2] = \underbrace{\text{Bias}^2}_{\text{underfitting}} + \underbrace{\text{Variance}}_{\text{overfitting}} + \underbrace{\sigma^2}_{\text{irreducible}}$$

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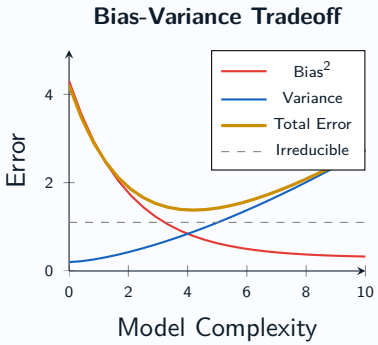

Diagnostic Tool

Compare **train error** vs **val error**.
Large train error ⇒ high bias.
Small train, large val ⇒ high variance.

L1 and L2 Regularization

L2 Regularization (Weight Decay)

Add penalty proportional to $\|\mathbf{w}\|_2^2$:

$$\mathcal{L}_{\text{reg}} = \mathcal{L} + 0.5\lambda \sum_j w_j^2$$

- Penalizes large weights
- Encourages **small, distributed** weights
- Smooth shrinkage toward zero

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Elastic Net

Combine both: $\mathcal{L}_{\text{reg}} = \mathcal{L} + \alpha\lambda\|\mathbf{w}\|_1 + \frac{(1-\alpha)\lambda}{2}\|\mathbf{w}\|_2^2$. Best of both worlds.

Dropout Regularization

Idea: Randomly **zero out** neurons during training with probability p.

For each unit j at training time:
$r_j \sim \text{Bernoulli}(1-p)$
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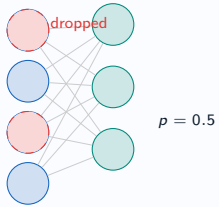

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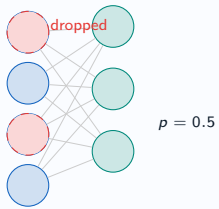

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Typical values

Dense layers: $p = 0.5$

Conv layers: $p = 0.1$–0.3

Input layer: $p = 0.2$

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Placement

Place BN **after** the linear transformation and **before** the activation. Common: Conv → BN → ReLU.

Early Stopping

Concept: Monitor validation loss and stop training when it stops improving.

- Simple and effective regularization
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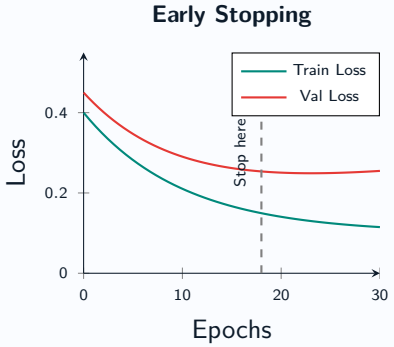

Pitfall

Don't use the test set for early stopping. Always use a separate **validation set**.

Data Augmentation

Idea: Artificially expand training set with label-preserving transformations.

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Image Augmentations:

- Random flip, crop, rotation
- Color jitter, brightness/contrast
- Cutout / Random Erasing
- Mixup: $\tilde{x} = \lambda x_i + (1 - \lambda) x_j$
- CutMix: paste patches between images

Hyperparameter Tuning

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Finding the best configuration for your model

What Are Hyperparameters?

Model Hyperparameters:

- Number of layers & units per layer
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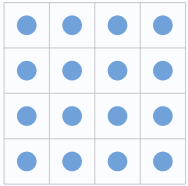
⚠ Key Principle

Hyperparameters are **not learned** during training. They must be set *before* training and validated on a **held-out validation set**.

Grid Search vs Random Search

Grid Search

- Exhaustively try all combinations
- Guaranteed to find best in grid
- Scales **exponentially** with # params
- Good for ≤ 2 hyperparameters



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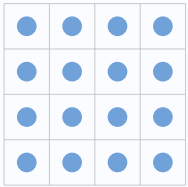
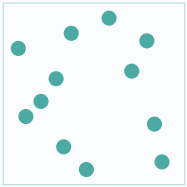


Grid Search (16 evals)

Random Search

- Randomly sample from distributions
- More efficient than grid (Bergstra & Bengio 2012)
- Explores **more unique values** per param
- Good for > 3 hyperparameters



Random Search (12 evals)

K-Fold Cross-Validation:

- Split data into K folds
- Train on $K - 1$, validate on 1
- Average metrics across all folds
- More reliable than single split

Cross-Validation & Model Selection

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Train / Val / Test Split:

Typical: 70% train / 15% val / 15% test
For large datasets: 98% / 1% / 1%

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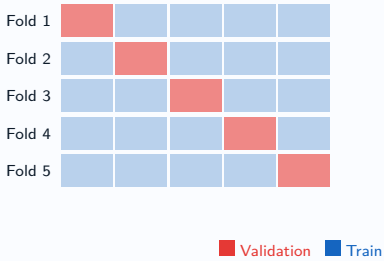


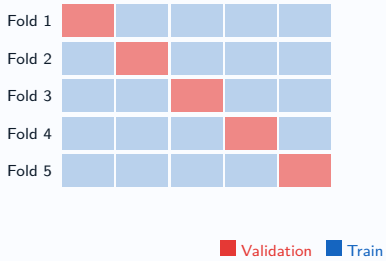
Train / Val / Test Split:

Typical: 70% train / 15% val / 15% test
For large datasets: 98% / 1% / 1%

Cross-Validation & Model Selection

K-Fold Cross-Validation:

- Split data into K folds
- Train on $K - 1$, validate on 1
- Average metrics across all folds
- More reliable than single split



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⚠ Data Leakage

Never use test set during model development or hyperparameter selection!

Weight Initialization

Why it matters:

- Bad init $\Rightarrow$ vanishing / exploding gradients
- Symmetry breaking is essential

Transfer Learning

Transfer Learning

Leveraging pretrained models for new tasks

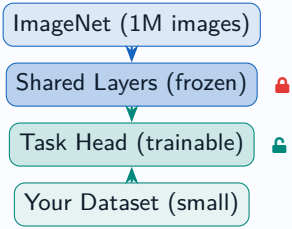
Transfer Learning: Concept

Motivation:

- Training large models from scratch is expensive
- Many tasks share common low-level features
- Pretrained models encode rich representations

Strategy:

1. Take a model **pretrained** on large dataset
2. **Freeze** the early layers (feature extractor)



Popular Pretrained Models

Computer Vision

- **ResNet-50/101/152** — residual networks
- **EfficientNet B0-B7** — accuracy/efficiency
- **ViT** — Vision Transformer
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Where to find them

- `huggingface.co/models` — 500,000+ pretrained models
- `pytorch.org/hub` — PyTorch Hub
- `tfhub.dev` — TensorFlow Hub

Overfitting and Small Datasets

Overfitting and Small Datasets

Generalization strategies when data is scarce

Diagnosing Overfitting

Signs of Overfitting:

- Training loss $\ll$ Validation loss
- High training accuracy, low val accuracy
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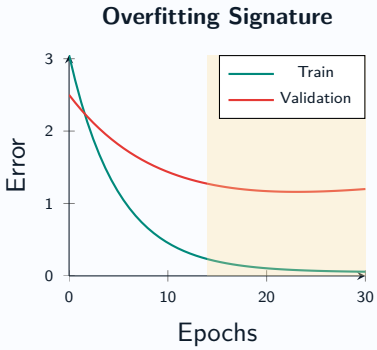

Strategies to Combat Overfitting

More Data

- Collect more labeled samples
- Data augmentation
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Training Strategies

- Early stopping
- Learning rate warmup
- Stochastic depth
- Label smoothing

Practical Deep Learning Checklist and Strategy

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- [x] Use appropriate weight init
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Golden Rule

Build a **simple baseline first**. Overfit intentionally on a tiny batch. Then scale up and regularize.

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✗ Mistake 6: No Validation Monitoring

Model overtrained without knowing.

Fix: Always use early stopping with

Summary: Techniques at a Glance

Technique	Purpose	When to Use	Key Param
Adam	Adaptive optimization	Default starting point	$\eta = 10^{-3}$
Dropout	Reduce overfitting	Dense layers	$p = 0.5$
BatchNorm	Stabilize training	After linear/conv	After layer
L2 / Weight Decay	Reduce overfitting	Any model	$\lambda = 10^{-4}$
Early Stopping	Prevent overtraining	All training	patience
Transfer Learning	Scarce data	Any domain	LR ratio
Data Augm.	Expand training set	Image/audio/text	aug policy
Label Smoothing	Regularization	Classification	$\epsilon = 0.1$